

2026 | Session 2 – COMP8240: Applications of Data Science

Assessment Task #1: Project Proposal

Submission details	
Due Date	11.55pm 28/08/26 Week 5
Weighting	Assessment will be marked out of a total of 20 marks. This assessment will contribute to 20% of overall unit grade.
Time to complete	This assessment will take approximately 20 hours to complete.
Length & Format	<i>Length:</i> A typical length would be 5-6 pages. <i>Material:</i> You will be expected to produce your report using latex (see week 3 material). You will submit a pdf that is the result of compiling the latex. <i>Style:</i> You should use the ACL template available in Overleaf. <i>References:</i> You should reference appropriately using the style available via the ACL template.
Late Penalties	Standard late penalty applies.
How to Submit	iLearn dropbox
Return of Assessment Grade & Feedback	Grades and feedback will be returned via iLearn, target timeframe 2 weeks from submission date.
Purpose	The overall purpose of the semester-long project is to develop and demonstrate capabilities in replicating a piece of research work, as is typically necessary in carrying out research involving computational experiments, using open source tools in industry, etc. In this specific assessment, you will develop a project proposal that introduces the selected work you intend to reproduce. The proposal should explain the data you will use, provide the necessary background, and outline a plan for carrying out the project.
Learning Outcomes Assessed	On successful completion you will be able to: • ULO01 Define and manage a project involving empirical research. • ULO03 Articulate clearly a coherent argument in written and oral form to a variety of audiences.
Skills assessed	Using the methods and tools described in Lectures and SGTAs, this task allows you to demonstrate: 1. your written communication skills; 2. your ability to recognise research that is likely to be reproducible, and to understand the core aspects of that paper that are involved in reproducibility; 3. your ability to define and manage a project involving empirical research; and 4. your ability to use tools typically used in the conduct and presentation of empirical computational research (e.g. Virtual Machines, LaTeX).
How will this task be assessed?	Individual

Use of AI	Open. Note that in using generative AI, you are responsible for its output, and for the relevant assessed aspects of the proposal document. For example, in this assessment you may use generative AI to suggest references, but you are expected to read all references to determine their relevance and correctness with respect to your topic.
Short automated extensions	Accepted.

Task overview

For this assessment, you will write a proposal document as outlined in the instructions below.

Task details / specific instructions

Goal

The goal of this report is to outline what you will do for your major project. For all projects, you will be (1) describing the paper you have chosen to work on, (2) explaining how the work in that paper is evaluated, and (3) explaining what data you will be using (both the data used in the original paper, and new data that you will find / construct) and where it will come from. If you are doing a novel project, in addition to the above, you will justify your choice of paper, with the goal of convincing us that it is suitable and feasible (and you will get feedback regarding this, with the option of falling back to the [default project](#) [this link is to the iLearn page describing these]).

The rest of this document should be read in conjunction with the [Overall Goal and Assessment Criteria](#) [this is similarly an iLearn link to the page of this name] for the project.

Report Structure

The report should have the following structure.

Description of Paper

You should describe what the paper is about and what it aims to achieve. This description should be aimed at a general computer science reader: someone who is familiar with computing in general, but does not necessarily know about the specific topic of the paper.

A good starting point is the abstract, introduction and conclusion of the paper itself. However, these will typically make some assumptions about the knowledge of the reader, who will normally already be familiar with the topic. For example, in the sample paper discussed in lectures on [Bayesian Unsupervised Topic Segmentation](#), the first sentence of the abstract says, "This paper describes a novel Bayesian approach to unsupervised topic segmentation." In your description of the paper, you would need to explain what unsupervised topic segmentation is. It can often be useful to provide an example, which you might take from your chosen paper or from another paper.

Explanation of Evaluation

A paper will typically make a lot of assumptions regarding the reader's knowledge of evaluation methodology and metrics. You will need to explain these in more detail, by reference to other papers (where the methodologies and metrics are defined, and possibly other papers where they are explained or clarified). For example, in the paper on [Bayesian Unsupervised Topic Segmentation](#), the metrics used include P_k and WindowDiff. You will need to explain these; this explanation should include a worked example.

As part of this, you should also describe the datasets used in the paper. You will typically need to give more detail than is available in the paper, and will need to refer to the original paper that describes the release of the dataset.

In addition, you will be introducing a new kind of evaluation, using an LLM as a judge. As we will only be discussing this in detail in lectures in week 7 and later, you do not need to have too much detail about how this will be done at this point.

Justification for Paper Choice [novel project only]

Check what's expected under [Overall Goal and Assessment Criteria](#). In addition to justifying the choice of work by some indicator of quality, you'll also need to convince us of the feasibility of the project. Is the code accessible and does it look likely to be able to be successfully implemented? Have a lot of people used it already (as measured by, e.g., number of github forks)? Is the original data actually available?

As a basic requirement, if you are proposing a new project, you should at least download and execute the relevant software, and note that you have done this successfully in this section of the report. You are not expected to go into technical details at this stage; that will be covered in the Update Presentation assessment. We simply expect a preliminary check to ensure feasibility. (If you cannot run the code, you should do the [default project](#).)

New Data

You should describe the new data that you will apply the method of the paper to. There should be two types of new data:

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1. Existing datasets that the method hasn't yet been applied to. You should discuss several (e.g., five) possibilities, and make an assessment of the feasibility of using each of them. You might find such datasets at places like Kaggle, HuggingFace, etc.
 2. New datasets that you create. For example, for [Bayesian Unsupervised Topic Segmentation](#), you might create your own datasets by splitting and concatenating Wikipedia articles. For many tasks, you may need some human judgements gathered. As we will only be discussing this in detail in lectures in week 5 and later, you do not need to have too much detail about how this will be done at this point.
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Quality Criteria

A high-quality submission:

- is well-structured with clear sections and logical flow;

- has an accurate characterisation of the paper to be investigated and reproduced;
- justifies the choice of paper (for novel projects only);
- has a clear explanation of the evaluation methodology in the chosen paper, along with some initial thoughts about a new evaluation using LLM-as-judge; and
- has a good description of both existing datasets and proposed new datasets.

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The rubric below guides you to what we expect.

Assessment Component	Outstanding High Distinction (100%-85%)	Advanced Distinction (84%-75%)	Proficient Credit (74%-65%)	Functional Pass (64%-50%)	Developing Fail (49%-0%)
Quality of description of chosen paper (4 marks)	Report very clearly describes purpose and relevant details of chosen paper with no significant errors.	Report does a good job of describing purpose and relevant details of chosen paper, possibly with an error or two.	Report is mostly accurate in describing purpose and relevant details of chosen paper.	Report conveys the general gist of the chosen paper.	Report does not successfully convey even the general gist of the chosen paper.
Justification of chosen paper (4 marks for novel project only; 0 marks if default project chosen)	Chosen paper is justified by several criteria, well explained.	Chosen paper is justified by several criteria, moderately well explained.	Chosen paper is justified by more than one criterion, with some explanation.	Chosen paper is justified in some (relevant) way.	Chosen paper has little or no justification.
Quality of explanation of evaluation aspects of chosen paper, including datasets used in that paper (4 marks)	Report explains all of the most important evaluation aspects of the chosen paper clearly, and describes all of the datasets used.	Report explains most of the most important evaluation aspects of the chosen paper clearly, and describes most of the datasets used.	Report explains some evaluation aspects of the chosen paper with some degree of clarity, and gives a reasonable description of the datasets used.	Report gives an overall sense of the evaluation and datasets of the chosen paper.	Report misses much of the explanation of evaluation and datasets used.
Quality of description of new data (4 marks)	Report explains very clearly and in some detail the specific new existing datasets to be used as well as the idea for newly constructed data.	Report explains the specific new existing datasets to be used, in less detail than expected for the HD level, as well as the idea for newly constructed data.	Report explains gives some idea for specific new existing datasets to be used as well as the idea for newly constructed data.	Report makes some suggestions for new existing datasets to be used, with less specificity than at the Cr level, as well as the idea for newly constructed data.	Report is missing most or all details on new datasets.
Quality of presentation (4 marks)	Writing style is excellent and clearly conveys the required content, and the report conforms to the specified format.	Writing style is very good and mostly conveys the required content, and the report mostly conforms to the specified format.	Writing style is decent and satisfactorily conveys the required content, and the report generally conforms to the specified format.	Writing style is satisfactory and goes some way to conveying the required content, and the report makes a reasonable attempt to conform to the specified format.	Writing style is unclear and/or the report fails to conform to the specified format in major ways.